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Multi-parameter imaging of dual-layer detector spectral CT in improving the imaging quality of gastrocolic trunk of henle (GTH)

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Abstract

Purpose To investigate the clinical application value of multi-parameter imaging with dual-layer detector spectral CT (DLCT) for GTH visualization.

Methods A total of 61 patients who underwent abdominal portal venous phase-enhanced SLCT were retrospectively enrolled. The raw image data were reconstructed into 120 kVp conventional images ($CI_{120\text{ kVp}}$) and virtual monoenergetic images (VMIs) at 40, 50, 60, and 70 keV ($VMI_{40-70\text{ keV}}$). Regions of interest (ROIs) were placed on the main trunk of GTH and the erector spinae muscle at the corresponding level to measure CT values and standard deviations (SD, indicative of image noise). The CT values, image noise, signal-to-noise ratios (SNRs), and contrast-to-noise ratios (CNRs) of VMI and $CI_{120\text{ kVp}}$ were analyzed. The image quality and anatomical information of each group were subjectively evaluated by integrating three-dimensional (3D) reconstruction technology, iodine density map, and effective atomic number (Z_{eff}) map.

Results For GTH visualized on $VMI_{40\text{ keV}}$, the CT value (530.56 ± 82.33 HU), SNR (25.34 ± 5.63), and CNR (57.85 ± 12.43) was significantly higher than those of the remaining groups (all $P < 0.001$). Subjective scoring indicated that $VMI_{40\text{ keV}}$ achieved the optimal image quality and vascular visualization performance. The iodine density map demonstrates relatively high density in vascular regions, the Z_{eff} map shows distinct coloration between vascular regions and non-vascular areas.

Conclusion $VMI_{40\text{ keV}}$ using DLCT provides better GTH image quality compared with $CI_{120\text{ kVp}}$, which may offer valuable support for clinical practice and preoperative surgical planning.

Keywords Dual-layer detector spectral CT (DLCT), Gastrocolic trunk of Henle, Virtual monoenergetic image

1 Introduction

GTH typically arises from the confluence of multiple tributaries, including the right gastroepiploic vein (RGEV), right colic vein (RCV), and anterior superior pancreaticoduodenal vein (ASPDV), occasionally, the superior right colic vein (SRCV) and middle colic



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vein (MCV) may also be incorporated. Characterized by a high rate of anatomical variation and a predisposition to intraoperative bleeding, the GTH serves as a critical control point in procedures such as laparoscopic right hemicolectomy [1]. Thus, accurate characterization and diagnosis of GTH variations are crucial for preventing iatrogenic vascular injury and ensuring radical tumor resection. To date, most GTH-related studies have relied on intraoperative observations and cadaveric dissections [2–4]. CT, one of the most widely used diagnostic modalities in clinical practice, enables clear, accurate, and non-invasive visualization of human vascular anatomy [5], thereby facilitating large-scale in vivo studies of the GTH. However, the dilution of intravascular contrast agent concentration during systemic circulation impairs the clear visualization of small branches (e.g., ASPDV) [6, 7], limiting the anatomical assessment of the GTH via CT venography. DLCT can optimize the contrast between vessels and surrounding tissues by extracting VMI, while enhancing the CT values of target tissues to achieve optimal image quality. While this technique has been applied in arterial imaging [8, 9], its application in GTH imaging remains unreported. In this study, we evaluated the imaging quality of the GTH using multi-parameter DLCT images combined with 3D reconstruction technology, with the aim of identifying the optimal VMI for GTH visualization. In summary, this is a novel imaging method that significantly improves the image quality of GTH. Our findings are expected to provide a theoretical basis and reference for clinical practice and precision surgery in abdominal surgery.

2 Materials and methods

2.1 Patients

A total of 94 patients who underwent abdominal enhanced spectral CT scans between August 2025 and September 2025 were retrospectively reviewed, with 61 patients ultimately included after meeting the study eligibility criteria. Among the included patients, there were 35 males and 26 females, aged 36–81 years, with a mean age of (60.91 ± 12.82) years. Inclusion criteria were defined as follows: (1) Visualization of the GTH in the portal venous phase of abdominal enhanced CT; (2) No prior gastrointestinal surgery before CT examination; (3) Without respiratory artifacts. Exclusion criteria included: (1) Severe CT image artifacts precluding diagnostic adequacy; (2) History of abdominal or vascular surgery; (3) Obscured GTH visualization due to adjacent tumors; (4) Severe hepatic or renal dysfunction (Fig. 1).

2.2 Computed tomography protocols and scanner

All patients underwent enhanced abdominal CT scan with SLCT (Spectral CT 7500, Philips Healthcare, the Netherlands). The patients were positioned in the supine position for scanning. The scanning range spanned from the hepatic dome level to the ischial tuberosity level. Bolus Tracking technology was employed for contrast agent monitoring, with the tracking plane set at the descending aorta and a trigger threshold of 110 HU. Arterial phase and portal venous phase scans were acquired at 10 s and 36 s post-threshold triggering, respectively. Scanning parameters were as follows: tube voltage: 120 kV, tube current: adaptive, pitch: 0.85, gantry rotation time: 0.75 s, matrix: 512×512 , slice thickness: 1 mm, slice interval: 1 mm. A non-ionic iodinated contrast agent, iopamidol (370 mgI/mL, Beijing Northland Pharmaceutical Co., Ltd.), was administered via the antecubital vein using a high-pressure injector (MEDRAD® Centargo CT Injection

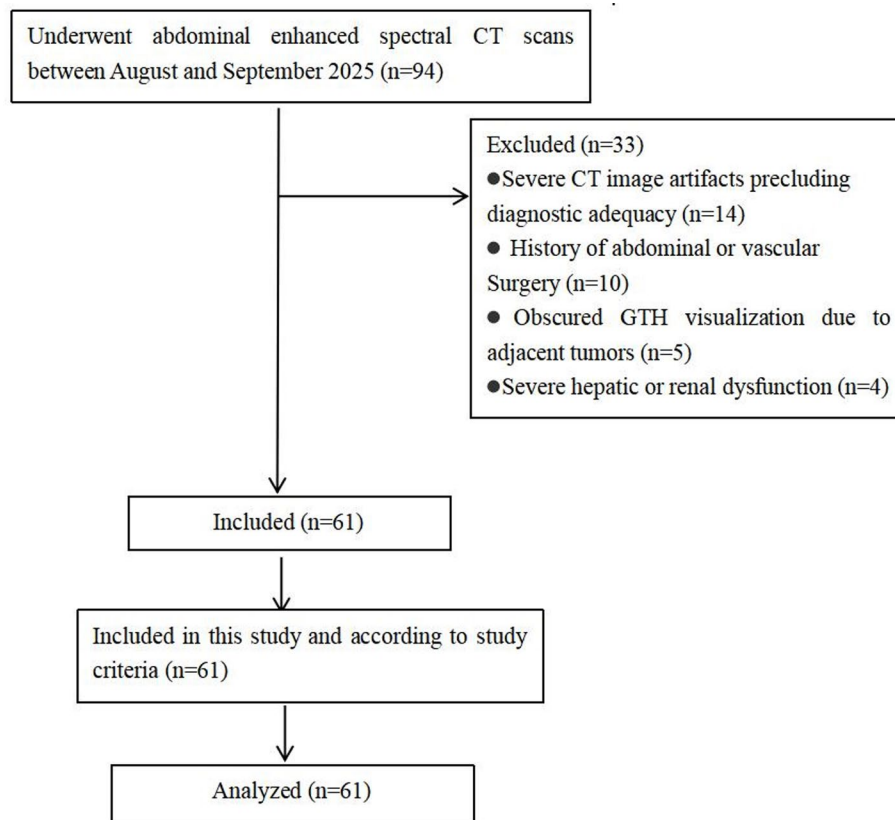


Fig. 1 Patient enrollment, inclusion, and exclusion

System, Bayer AG, Germany) at an infusion rate of 3.0 mL/s, with a total dose ranging from 1.5 to 2.0 mL/kg of body weight. A spectral-based imaging (SBI) dataset was automatically generated post-acquisition.

2.3 Image reconstruction and analysis

The SBI data were transferred to the Philips IntelliSpace Portal workstation, $CI_{120\text{ kVp}}$, $VMI_{40-70\text{ keV}}$ iodine density maps, and Zeff maps were reconstructed. Subsequently, volume rendering (VR) images were generated based on the $CI_{120\text{ kVp}}$ and $VMI_{40-70\text{ keV}}$. ROIs were placed on the main trunk of GTH and the left erector spinae muscle at the same axial level. The CT values and SD were measured, followed by the calculation of SNR and CNR. To ensure consistency, the diameter of each ROI was set to two-thirds of the GTH lumen diameter, and vascular walls were carefully avoided. All measurements were performed independently by two senior radiologists, and the mean values were adopted for subsequent analyses. A schematic illustration of the measurement protocol is provided in Fig. 2.

The calculation formulas for SNR and CNR are as follows:

$$\text{SNR} = \text{CT value}_{(\text{GTH})} / \text{SD}_{(\text{GTH})}$$

$$\text{CNR} = [\text{CT value}_{(\text{GTH})} - \text{CT value}_{(\text{erector spinae muscle})}] / \text{SD}_{(\text{erector spinae muscle})}$$

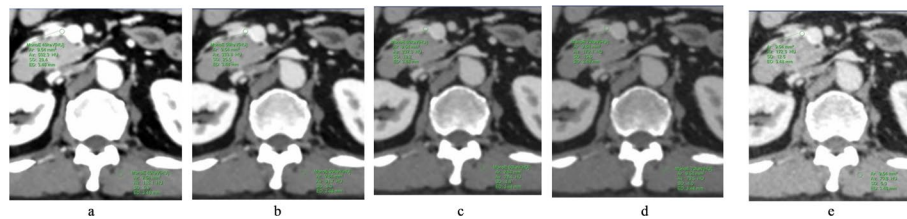


Fig. 2 CT values and SD of the GTH and erector spinae muscle at the same axial level in the same patient, measured at VMI_{40 keV} (a), VMI_{50 keV} (b), VMI_{60 keV} (c), VMI_{70 keV} (d), and CI_{120 kVp} (e)

Subjective evaluation of VR images were independently performed by two senior radiologists, with the scoring criteria defined as follows: 5 points: excellent vessel signal intensity with very sharp edge of vessels; 4 points: good signal intensity with sharp edge of vessels; 3 points: fair signal intensity with fairly sharp edge of vessels; 2 points: bad signal intensity with unclear vessel edge; 1 points: worse signal intensity with blurred edge of vessels [10].

2.4 Statistical analysis

The SPSS 22.0 software (SPSS, Chicago, IL, USA) was used for all statistical analyses. All parameter values were tested for normality and homogeneity of variance. Data conforming to a normal distribution were expressed as the mean $\pm$ standard deviation. Inter-group comparisons were performed using one-way analysis of variance (ANOVA), followed by Tukey's test for post-hoc pairwise comparisons. For data deviating from a normal distribution, values were expressed as median (first quartile, third quartile) [M (Q1, Q3)], with inter-group comparisons conducted via the Kruskal-Wallis H test and Bonferroni correction applied for pairwise comparisons. The inter-rater reliability of subjective image quality scores between the two radiologists was assessed using the Kappa test. $P < 0.05$ was set as the statistical significance threshold.

3 Results

The CT values: No statistically significant differences were observed between the CI_{120 kVp} and VMI_{70 keV} groups ($P > 0.05$), whereas statistically significant differences were noted among all other groups ($P < 0.001$). The CT value of VMI_{40 keV} was the highest, with increasing energy levels, the CT values of the vessels gradually decreased.

Image noise: No statistically significant differences were observed between the CI_{120 kVp} and VMI_{70 keV} groups ($P > 0.05$), whereas statistically significant differences were noted among all other groups ($P < 0.001$). The VMI_{40 keV} group demonstrated the highest image noise, with increasing energy levels, the image noises gradually decreased across all groups.

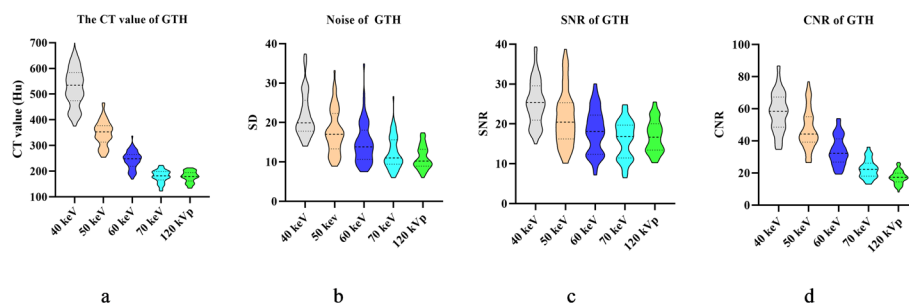
SNR: No statistically significant differences were observed among the VMI_{60 keV}, VMI_{70 keV}, and CI_{120 kVp} groups ($P > 0.05$), whereas statistically significant differences were noted among all other groups ($P < 0.001$), with increasing energy levels, the SNR of each group gradually decreased, and the VMI_{40 keV} group exhibited the highest SNR.

CNR: No statistically significant differences were observed between the VMI_{70 keV} and CI_{120 kVp} groups ($P > 0.05$), whereas statistically significant differences were noted among all other groups ($P < 0.001$), with increasing energy levels, the CNR of each group gradually decreased, and the VMI_{40 keV} group exhibited the highest CNR.

Table 1 Comparison of CT values, noise, SNR, and CNR among different image groups

Group	CT value	Noise	SNR	CNR
40 keV	530.56 ± 82.33	19.90 (17.80, 25.65)	25.34 ± 5.63	57.85 ± 12.43
50 keV	346.35 ± 48.44	17.00 (13.30, 22.30)	21.46 ± 6.92	47.09 ± 12.37
60 keV	243.66 ± 34.41	13.80 (10.65, 18.05)	18.14 ± 5.52	40.00 ± 9.01
70 keV	179.81 ± 23.70	11.00 (9.40, 15.60)	16.03 ± 4.79	22.65 ± 5.79
120 kVp	178.08 ± 21.33	10.20 (8.90, 13.20)	16.98 ± 4.08	17.24 ± 4.01
<i>P</i>	< 0.001	< 0.001	< 0.001	< 0.001
<i>P</i> *	< 0.001	0.006	0.205	< 0.001
<i>P</i> **	< 0.001	< 0.001	0.761	< 0.001
<i>P</i> ***	0.100	0.114	0.874	0.013

P: 40 keV vs. 50 keV, 40 keV vs. 60 keV, 40 keV vs. 70 keV, 40 keV vs. 120 kVp, 50 keV vs. 60 keV, 50 keV vs. 70 keV, 50 keV vs. 120 kVp; *P**: 60 keV vs. 70 keV; *P***: 60 keV vs. 120 kVp; *P****: 70 keV vs. 120 kVp

**Fig. 3** CT values (a), noise (b), SNR (c), and CNR (d) of images from different groups**Table 2** Inter-rater consistency of subjective image evaluation among different groups

Doctor	40 keV	50 keV	60 keV	70 keV	120 kVp
A	4.92 ± 0.12	4.69 ± 0.09	4.09 ± 0.06	3.21 ± 0.26	3.22 ± 0.26
B	4.95 ± 0.10	4.67 ± 0.11	4.10 ± 0.06	3.20 ± 0.27	3.21 ± 0.25
Kappa	0.90	0.93	0.89	0.94	0.92
95% CI	0.66–1.00	0.87–0.98	0.82–0.96	0.88–0.97	0.85–0.98

The CT values, noises, SNRs, and CNRs of the $CI_{120\text{ kVp}}$ and $VMI_{40-70\text{ keV}}$ groups are presented in Table 1; Fig. 3.

The subjective image quality evaluation: The inter-rater consistency between the two radiologists was excellent ($Kappa \geq 0.89$), as presented in Table 2. No statistically significant differences were observed in subjective scores between the $VMI_{70\text{ keV}}$ and $CI_{120\text{ kVp}}$ groups ($P = 0.805$), whereas statistically significant differences were noted among all other groups ($P < 0.001$), with increasing energy levels, the subjective scores gradually decreased. The $VMI_{40\text{ keV}}$ group exhibited the highest subjective score, with clear visualization of the anatomical structure and details of the GTH, as well as good contrast with surrounding tissues (Fig. 4a–e).

Iodine density map: The vascular region exhibited high-density signals, whereas other regions showed relatively low-density signals (Fig. 5a); Zeff map: The vascular region and non-vascular regions displayed distinct coloration (Fig. 5b).

4 Discussion

In 1868, anatomist Henle first described the common trunk structure formed by RGEV and RCV, which was subsequently termed the GTH [11]. With the widespread adoption of radical resection for right-sided colon cancer, the GTH has emerged as a core

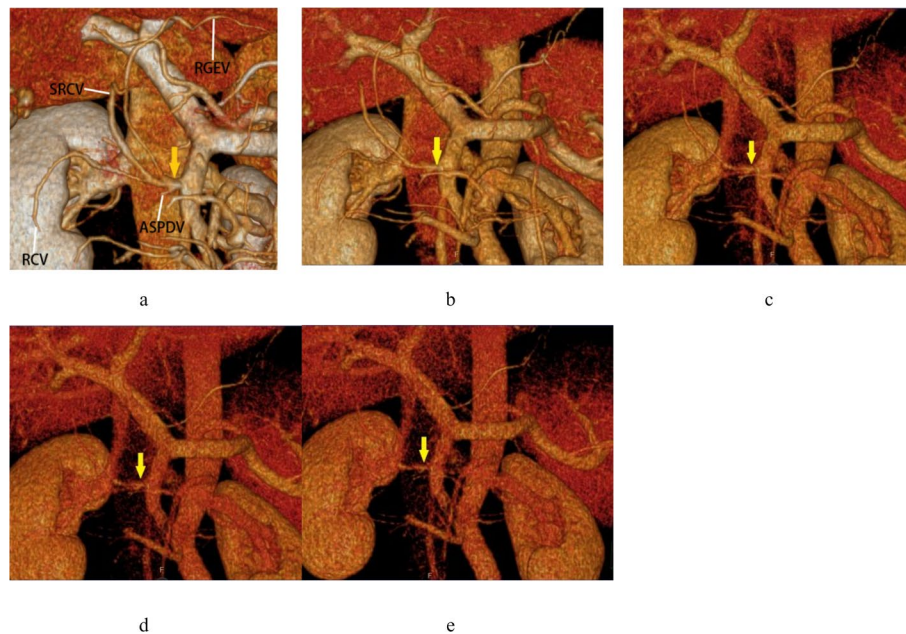


Fig. 4 Volume rendering (VR) images of the same patient at 40 keV (a), 50 keV (b), 60 keV (c), 70 keV (d), and 120 kVp (e). The arrow points to the GTH

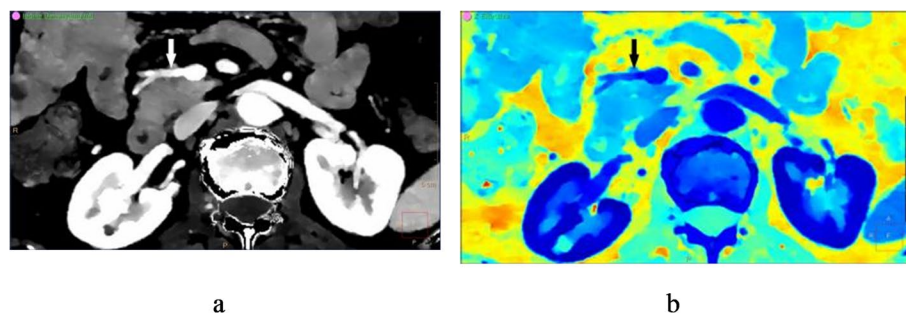


Fig. 5 Iodine density map (a) and Zeff map (b). The arrow points to the GTH

landmark for delineating the surgical plane, guiding vascular manipulation, and assessing the radicality of resection. Owing to its deep anatomical location, variable tributary patterns, and short, robust main trunk, iatrogenic injury during surgery frequently results in refractory venous hemorrhage. In recent years, advances in three-dimensional (3D) imaging have facilitated the maturation of anatomical insights, preoperative evaluation protocols, and surgical management strategies for the GTH [12].

Conventional CT exhibits limitations in visualizing small venous vessels, primarily attributed to the reduced contrast agent concentration within the portal vein. DLCT employs a stacked dual-detector configuration: the upper detector layer absorbs low-energy X-ray photons, while the lower layer captures high-energy photons. This design enables simultaneous dual-energy data acquisition and decoupling in a single scan, eliminating the need for repeated scanning or voltage switching. Retrospective generation of multimodal parameters, including $VMI_{40-200\text{ keV}}$, iodine density map, and Zeff map—is feasible without additional radiation exposure [13, 14].

In this study, the CT values of GTH were compared across $VMI_{40-70\text{ keV}}$ and $CI_{120\text{ kVp}}$ group. The results demonstrated that, with the exception of $VMI_{70\text{ keV}}$, the CT values

of GTH in all other VMI groups were significantly higher than those in the $CI_{120\text{ kVp}}$ group—indicating that spectral CT can enhance vascular enhancement. Among these groups, the $VMI_{40\text{ keV}}$ yielded the most pronounced elevation in CT value of GTH, which is consistent with related vascular studies [15, 16]. Low-energy VMI (40–60 keV) approximates the K-edge of iodine (33.2 keV), markedly amplifying the photoelectric effect, this leads to a significant increase in the CT values of iodine-containing vessels and facilitates the visualization of distal small vessels [17].

Image noise exhibited a downward trend with increasing energy levels. Among the groups, $VMI_{40\text{ keV}}$ showed the highest noise, while $CI_{120\text{ kVp}}$ had the lowest noise—consistent with prior reports [18]. This phenomenon can be attributed to two key factors: first, low-energy X-rays have limited penetration, leading to insufficient photon flux detected by the detector and thus marked amplification of quantum noise; second, $VMI_{40\text{ keV}}$ yield the highest venous CT values, which increases the degree of inhomogeneity and further contributes to elevated noise levels [19, 20]. Studies focusing on lower extremity veins have reported that the image noise of $VMI_{40\text{ keV}}$ is lower than that of $CI_{120\text{ kVp}}$ [21], which is inconsistent with this study. This discrepancy can be attributed to the relatively small caliber of GTH, which renders it more susceptible to partial volume effects and uneven contrast agent filling, consequently, the image noise of GTH is higher than that of large-caliber vessels such as the portal vein, aorta, and lower extremity vessels [22–24]. Nonetheless, the noise level of $VMI_{40\text{ keV}}$ remains less than twice that of $CI_{120\text{ kVp}}$, this outcome is facilitated by the anti-correlation noise suppression technology integrated into SLCT, which not only mitigates noise but also fully preserves the high-contrast advantage of $VMI_{40\text{ keV}}$. In contrast, no significant difference in image noise was observed between $VMI_{70\text{ keV}}$ and $CI_{120\text{ kVp}}$. This indicates that $VMI_{70\text{ keV}}$ benefits from a higher photon flux and more uniform enhancement, leading to inherently reduced noise and greater similarity to $CI_{120\text{ kVp}}$. Despite exhibiting the highest noise level, $VMI_{40\text{ keV}}$ achieved the optimal SNR and CNR, with statistically significant differences compared to $VMI_{50-70\text{ keV}}$ and $CI_{120\text{ kVp}}$. Additionally, $VMI_{40\text{ keV}}$ received the highest subjective image quality scores, confirming it as the optimal imaging approach for visualizing the GTH. Prior studies have established that low-energy VMI enhances the imaging quality of the common hepatic artery, portal vein, and hepatic vein [25], further, $VMI_{40\text{ keV}}$ has been shown to deliver the highest diagnostic performance for characterizing portal vein thrombosis and to provide the most favorable visualization of the adrenal vein [26, 27], which is consistent with this study. Regarding clinical workflow, Yoshida et al. compared $VMI_{40\text{ keV}}$ of the portal vein with CI in the late venous phase [27], which is more refined. However, it is impractical to intentionally scan the late venous phase in enhanced abdominal CT. Furthermore, adrenal veins are tributaries of the inferior vena cava, while GTH is a tributary of the portal vein, so they are not comparable. Some studies have placed the ROI on the psoas major muscle as background noise [28, 29], yet using the erector spinae muscle as background noise is more common in academic research [15, 22, 30]. Notably, neither approach has been shown to exert a significant impact on the study outcomes.

In iodine density map, iodine-free voxels are calibrated to 0 mg/ml (displayed as black), while iodine-containing tissues are visualized in grayscale corresponding to their iodine concentrations [31]. Iodine density map [mg/ml] hold the potential to quantify iodine enhancement and improve the visualization of iodine within contrast-enhanced tissues,

thereby enhancing the detectability of microvessels. Zeff map facilitate more prominent vessel visualization through pseudocolor staining of distinct substances. However, iodine density map and Zeff map cannot be subjected to 3D reconstruction, precluding stereoscopic assessment of vessels. As such, they can serve as a robust complement to $VMI_{40\text{ keV}}$.

This study has several limitations. First, as a retrospective single-center study, it is susceptible to selection bias, which may limit the generalizability of the findings. Second, the anatomical classification of GTH based on $VMI_{40\text{ keV}}$ requires validation with a larger sample size to enhance the robustness of the results. Third, further investigations are warranted to explore whether optimizing contrast agent parameters (e.g., concentration, dose, and injection rate) can further improve the imaging quality of GTH, as well as to determine whether low-voltage, low-radiation-dose scanning protocols can serve as a viable alternative to $CI_{120\text{ kVp}}$.

In conclusion, $VMI_{40\text{ keV}}$ with DLCT exhibits the optimal vascular optimization effect for the GTH, significantly enhancing vascular imaging quality. When combined with iodine density map and Zeff map, the branching patterns of the GTH can be visualized more comprehensively. With the continuous advancement of CT and scanning technologies, as well as the deepening of research on the GTH, it is believed that DLCT venous imaging technology will be increasingly widely applied in clinical practice in the future.

Author contributions

Protocol/project development: X He and Y Pan; Data collection or management: J Xiao, N Mu and C Wang; Data analysis: J Li and L Liu; Manuscript writing/editing: X He and Y Shan.

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Data availability

The datasets analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Review Committee of Rizhao People's Hospital, Shandong Province (Approval No.: 2024-Ethics Opinion-MR-125-01; Date: January 6, 2025) and complies with the Declaration of Helsinki and its subsequent revisions. Informed consent was obtained from all individual participants included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. He Z, Su H, Ye K, et al. Anatomical characteristics and classifications of gastrocolic trunk of Henle in laparoscopic right colectomy: preliminary results of multicenter observational study. *Surg Endosc.* 2020;34:4655–61. <https://doi.org/10.1007/s00464-019-07247-2>.
2. He Z, Yang C, Diao D, et al. Anatomic patterns and clinical significance of gastrocolic trunk of Henlé in laparoscopic right colectomy for colon cancer: Results of the HeLaRC trial. *Int J Surg.* 2022;104:106718. <https://doi.org/10.1016/j.jssu.2022.106718>.
3. Ghimire R, Thapa N, Acharya BP, et al. Intraoperative variations of the gastrocolic trunk of henle noted in gastrointestinal surgeries. *J Nepal Health Res Counc.* 2024;22:366–9. <https://doi.org/10.33314/jnhrc.v22i02.5132>.
4. Stefura T, Kacprzyk A, Droś J, Pędziwiatr M, Major P, Holda MK. The venous trunk of henle (gastrocolic trunk): a systematic review and meta-analysis of its prevalence, dimensions, and tributary variations. *Clin Anat.* 2018;31:1109–21. <https://doi.org/10.1002/ca.23228>.
5. Tang W, Shi J, Kuang LQ, Tang SY, Wang Y. Celiomesenteric trunk: New classification based on multidetector computed tomography angiographic findings and probable embryological mechanisms. *World J Clin Cases.* 2019;7:3980–9. <https://doi.org/10.12998/wjcc.v7.i23.3980>.

6. Obmann MM, Gehweiler J, Schindera ST, Janetzki J, Boll DT, Benz MR. Clinical evaluation of a novel multibolus contrast agent injection protocol for thoraco-abdominal CT angiography: assessment of homogeneity of arterial contrast enhancement. *Eur J Radiol.* 2020;126:108957. <https://doi.org/10.1016/j.ejrad.2020.108957>.
7. Wu R, Watanabe Y, Satoh K, et al. Quantitative comparison of virtual monochromatic images of dual energy computed tomography systems: beam hardening artifact correction and variance in computed tomography numbers: a phantom study. *J Comput Assist Tomogr.* 2018;42:648–54. <https://doi.org/10.1097/RCT.0000000000000726>.
8. Ferrández-Ferrández D, Arenas-Jiménez JJ, Ureña Vacas A, et al. Vascular attenuation and volumetric lung iodine density in dual-layer spectral CT pulmonary angiography: a randomized controlled trial comparing three contrast doses. *Eur Radiol.* 2026. <https://doi.org/10.1007/s00330-025-12309-2>.
9. Tan C, Gui S, Zhong Z, Chen X. The diagnostic value of second-generation dual-layer spectral detector CT for Stanford type A aortic dissection. *Eur Heart J Cardiovasc Imaging.* 2024;25:e261. <https://doi.org/10.1093/ehjci/jeae156>.
10. He J, Ma X, Wang Q, Fan J, Sun Z. Spectral CT demonstration of the superior mesenteric artery: comparison of monochromatic and polychromatic imaging. *Acad Radiol.* 2014;21:364–8. <https://doi.org/10.1016/j.acra.2013.11.004>.
11. Voiglio EJ, Bouillier du Retail C, Neidhardt JP, Caillot JL, Barale F, Mertens P. Gastrocolic vein. Definition and report of two cases of avulsion. *Surg Radiol Anat.* 1998;20:197–201. <https://doi.org/10.1007/s00276-998-0197-9>.
12. Zhao X, Zhang H, Zhao H, Kong D, Zeng W, Gao F, Meng X, Ke J. Anatomic association between the gastrocolic trunk of Henle and right colic artery by high-quality CT venography. *Sci Rep.* 2024;14:32054. <https://doi.org/10.1038/s41598-024-83588-w>.
13. Rassouli N, Etesami M, Dhanantwari A, Rajiah P. Detector-based spectral CT with a novel dual-layer technology: principles and applications. *Insights Imaging.* 2017;8:589–98. <https://doi.org/10.1007/s13244-017-0571-4>.
14. So A, Nicolaou S. Spectral computed tomography: fundamental principles and recent developments. *Korean J Radiol.* 2021;22:86–96. <https://doi.org/10.3348/kjr.2020.0144>.
15. Lai LY, Tan P, Jiang Y, Wang F, Dong J, Huang MP, Shu J. Dual-layer spectral detector CT for contrast agent concentration, dose and injection rate reduction: utility in imaging of the superior mesenteric artery. *Eur J Radiol.* 2022;150:110246. <https://doi.org/10.1016/j.ejrad.2022.110246>.
16. Chen Y, Liu X, Zeng H, Zhang J, Li Z, Wu B, Huang Z, Song B. The clinical applications of dual-layer spectral detector CT in digestive system diseases. *Eur Radiol.* 2025;35:3547–57. <https://doi.org/10.1007/s00330-024-11290-6>.
17. Rajiah P, Abbara S, Halliburton SS. Spectral detector CT for cardiovascular applications. *Diagn Interv Radiol.* 2017;23:187–93. <https://doi.org/10.5152/dir.2016.16255>.
18. Sakabe D, Funama Y, Taguchi K, Nakaura T, Utsunomiya D, Oda S, Kidoh M, Nagayama Y, Yamashita Y. Image quality characteristics for virtual monoenergetic images using dual-layer spectral detector CT: Comparison with conventional tube-voltage images. *Phys Med.* 2018;49:5–10. <https://doi.org/10.1016/j.ejmp.2018.04.388>.
19. Kalisz K, Rassouli N, Dhanantwari A, Jordan D, Rajiah P. Noise characteristics of virtual monoenergetic images from a novel detector-based spectral CT scanner. *Eur J Radiol.* 2018;98:118–25. <https://doi.org/10.1016/j.ejrad.2017.11.005>.
20. Martin SS, Albrecht MH, Wichmann JL, et al. Value of a noise-optimized virtual monoenergetic reconstruction technique in dual-energy CT for planning of transcatheter aortic valve replacement. *Eur Radiol.* 2017;27:705–14. <https://doi.org/10.1007/s00330-016-4422-3>.
21. Tanoue S, Nakaura T, Iyama Y, Iyama A, Nagayama Y, Yoshida M, Yamashita Y. Diagnostic performance of dual-layer computed tomography for deep vein thrombosis in indirect computed tomography venography. *Circ J.* 2020;84:636–41. <https://doi.org/10.1253/circj.CJ-19>.
22. Mangold D, Salatzki J, Riffel J, Kauczor HU, Weber TF. Dual-layer spectral CTA for TAVI planning using a split-phase protocol and low-keV virtual monoenergetic images: improved image quality in comparison with single-phase conventional CTA. *Rofo.* 2022;194:652–9. <https://doi.org/10.1055/a-1717>.
23. Ren H, Zhen Y, Gong Z, Wang C, Chang Z, Zheng J. Feasibility of low-dose contrast media in run-off CT angiography on dual-layer spectral detector CT. *Quant Imaging Med Surg.* 2021;11:1796–804. <https://doi.org/10.21037/qims-20-925>.
24. Lennartz S, Große Hokamp N, Zäske C, Zopfs D, Bratke G, Glauner A, Maintz D, Persigehl T, Chang DH, Hickethier T. Virtual monoenergetic images preserve diagnostic assessability in contrast media reduced abdominal spectral detector CT. *Br J Radiol.* 2020;93:20200340. <https://doi.org/10.1259/bjr.20200340>.
25. Lai LY, Jiang Y, Shu J. The application of dual-layer spectral detector CT in abdominal vascular imaging. *Curr Med Imaging.* 2023;19:1609–15. <https://doi.org/10.2174/1573405619666230216122650>.
26. Zhang L, Zheng T, Su M, Zhang X, Liu H, Song B, Chen Y. Dual-layer spectral detector computed tomography in distinguishing between bland and neoplastic portal vein thrombosis. *Quant Imaging Med Surg.* 2025;15:8850–63. <https://doi.org/10.21037/qims-24-1890>.
27. Yoshida R, Nagayama Y, Esaki T, et al. Dual-layer spectral detector CT for improved assessment of right adrenal vein in patients with primary aldosteronism. *Abdom Radiol (NY).* 2025;51:1974–85. <https://doi.org/10.1007/s00261-025-05165-7>.
28. Yin XP, Gao BL, Li CY, et al. Optimal monochromatic imaging of spectral computed tomography potentially improves the quality of hepatic vascular imaging. *Korean J Radiol.* 2018;19:578–84. <https://doi.org/10.3348/kjr.2018.19.4.578>.
29. Martin SS, Kolaneci J, Czwikla R, et al. Dual-energy CT for the detection of portal vein thrombosis: improved diagnostic performance using virtual monoenergetic reconstructions. *Diagnostics (Basel).* 2022;12:1682. <https://doi.org/10.3390/diagnostics12071682>.
30. Zhang X, Chen J, Yu N, et al. Reducing contrast medium dose with low photon energy images in renal dual-energy spectral CT angiography and adaptive statistical iterative reconstruction (ASIR). *Br J Radiol.* 2021;94:20200974. <https://doi.org/10.1259/bjr.20200974>.
31. Rassouli N, Etesami M, Dhanantwari A, Rajiah P. Detector-based spectral CT with a novel dual-layer technology: principles and applications. *Insights Imaging.* 2017;8:589–98.

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